



Decision Theory

Daniel C



Roadmap

From classical choice to embedded counterfactuals

- **Why decision theory matters for AI safety** (modelling, specification, reflective stability, multi-agent conflict)
- **Why it is hard:** optimisation is solved for *dualistic* agents but not for *embedded* ones, where the agent must *construct* counterfactuals
- **The candidates:** causal (CDT), evidential (EDT), functional/logical (FDT), updateless (UDT 1.0, UDT 1.1) – each is a different recipe for the counterfactual
- **The hard core:** problems with UDT, the 5-and-10 problem, logical counterfactuals and Löb's theorem
- **A unifying goal:** formalise decision problems as programs and ask which decision theory is *optimal* across all “fair” problems



Why study decision theory? (for AI safety)

Four reasons

- **Architecture-independent modelling.** We want to reason about how a superintelligent agent will *behave* without knowing the details of how it is built. Decision theory is a minimal model of "what is rational choice", capturing behaviour any sufficiently capable agent tends to converge on.
- **Implementation / the specification-optimisation split.** Alignment decomposes into (1) *specify* a goal we want, and (2) *optimise* it correctly. Decision theory is part (2). A perfect specification is **not sufficient** for safety: if the agent constructs the wrong counterfactuals, it can "optimise" the goal in a way we never intended.
- **Reflective stability.** Safety properties must remain invariant under self-modification (cf. Agent Foundations deck). Decision theory is the simplest setting to ask: *would an agent running decision theory A want to self-modify into, or build a successor with, decision theory B?* A decision theory that wants to overwrite itself is a warning sign.
- **Multi-agent dynamics.** Decision problems get hard under strategic interaction and mutual modelling. We want a decision theory under which agents can *cooperate* with other superintelligences and *avoid catastrophic conflict*, without being *exploitable*.



The easy case: dualistic agents

When the agent is cleanly separated from the world

- A *dualistic* agent has clean input/output channels with its environment (the “cybernetic” picture: a player optimising a video game). It is “**larger than**” the world (can hold a full model of it) and “**outside**” it (need not reason about itself).
- Here the action is a genuine *free variable*: nothing in the world depends on the agent’s internal computation except through the downstream consequences of the action. Choosing optimally is conceptually solved:

$$a^* = \arg \max_a \mathbb{E}[U \mid \text{do}(A = a)].$$

- This is the mature, successful framework of **expected-utility maximisation**: the von Neumann–Morgenstern / Savage axioms (preferences satisfying coherence are EU-maximisation), Bayesian updating, and **complete-class theorems** (every admissible decision rule is Bayes-optimal for some prior).
- It underpins modern economics, statistics, and reinforcement learning. The conceptual work is done; only computation remains.



The hard case: embedded agents

When the agent is a part of the world it reasons about

- An *embedded* agent is a piece of the world it is trying to optimise (cf. Agent Foundations deck). It can be copied, predicted, simulated; its source code can be read by others.
- Now **“which action I take” is just another fact about the world**, fixed by physics (or by the agent's own deterministic computation). There is no free variable to set, and **no ready-made answer to “what would happen if I acted differently”** – the alternative may be physically or logically impossible.
- So the counterfactual must be *constructed*, not read off. **This is the whole difficulty**: decision theory for embedded agents is the problem of constructing counterfactuals.
- Each decision theory is a different construction:
 - **EDT**: condition on the action as *evidence*, $\mathbb{E}[U \mid A = a]$
 - **CDT**: *intervene* causally on the action, $\mathbb{E}[U \mid \text{do}(A = a)]$
 - **FDT/UDT**: reason about the *logical output* of the decision procedure being a



A 60-second primer: Bayesian networks

Encoding “what depends on what”

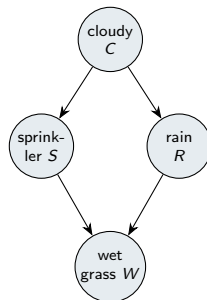
- A Bayesian network is a directed acyclic graph: nodes are random variables, an edge $X \rightarrow Y$ means Y *directly listens to* X . The joint distribution *factorises* along the graph:

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Pa}(X_i)),$$

with $\text{Pa}(X_i)$ the parents (direct causes) of X_i .

- Each variable is independent of its non-descendants given its parents: a compact, structured stand-in for a huge joint distribution.
- For the graph on the right:

$$P(C) P(S \mid C) P(R \mid C) P(W \mid S, R).$$



*cloud cover is a common cause of sprinkler use
and rain*



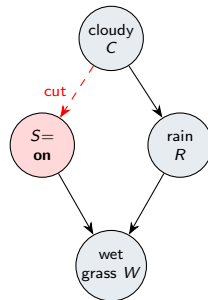
Causal decision theory: seeing vs. doing

The do-operator

- Pearl's key distinction: *observing* $X=x$ (passive evidence) is not the same as *doing* $X=x$ (an intervention).
- $\text{do}(X = x)$: **sever all arrows into** X (cut X off from its causes), fix $X = x$, propagate downstream. Adjusting over the direct causes $z = \text{Pa}(X)$:

$$P(Y \mid \text{do}(X=x)) = \sum_z P(Y \mid x, z) P(z).$$

- Sprinkler graph: *seeing* the sprinkler on raises the chance it was cloudy, hence lowers the chance of rain. *Doing* $\text{do}(S=\text{on})$ cuts $C \rightarrow S$, so it tells you **nothing** about rain.
- CDT**: $a^* = \arg \max_a \mathbb{E}[U \mid \text{do}(A = a)]$ – judge each action by the causal consequences of *intervening* on it, holding the rest of the past fixed.



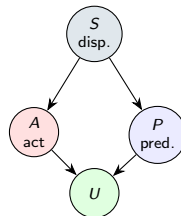
$\text{do}(S=\text{on})$ severs S from its cause C



CDT on Newcomb's problem

Two-boxing, and why

- **Newcomb:** a transparent box holds \$1,000; an opaque box holds \$1,000,000 *iff* a highly reliable predictor Ω foresaw you would take *only* the opaque box. Ω has already chosen.
- CDT applies $\text{do}(A)$: this **severs** your action from your disposition S , so it no longer carries information about Ω 's prediction P . The box contents are now fixed; taking both boxes earns \$1,000 more *in every fixed state*. CDT **two-boxes**.
- But predictably: two-boxers are predicted to two-box and walk away with \$1,000; one-boxers get \$1,000,000.
- CDT wins the battle (local dominance) and loses the war (the predictable policy is poor). It *deliberately discards* the dependence of Ω 's prediction on the agent's decision.



CDT cuts $S \rightarrow P$, losing the link to P .



CDT vs. physics: the intervention is a fiction

Does CDT really “reflect physical reality”? (following Ebtekar & Hutter)

- CDT's $\text{do}(A = a)$ holds the past fixed, **magically sets the action**, and propagates forward. What licenses cutting the action loose from its causes?
- In a *deterministic, time-reversible* physical world, your action has physical antecedents (your brain state, your source code). A predictor that reads those antecedents is correlated with your action through **perfectly ordinary physics**, not magic.
- “Severing the incoming arrows to *A*” directly **contradicts embeddedness**: it pretends your action has no causes, but physically it does. The intervention is a harmless idealisation for a dualistic agent; for an embedded agent it *throws away real correlations* (with predictors, copies, simulations).
- So the usual defence is backwards: physics gives no privileged metaphysical “intervention” that frees the action from its past. This pressures us to move the counterfactual off the *physical act* and onto the agent's *policy / computation*.



Evidential decision theory: condition, don't intervene

EDT as conditioning

- EDT evaluates an action by the *news* it would bring – it **conditions** on “I do a ” as evidence:

$$a^* = \arg \max_a \mathbb{E}[U \mid A = a].$$

- On **Newcomb**: one-boxing is strong evidence the opaque box is full (the predictor is reliable), so $\mathbb{E}[U \mid A = \text{one-box}]$ is huge. EDT **one-boxes** and gets the million. EDT captures exactly the predictive dependence CDT discards.
- The contrast in one line: $\mathbb{E}[U \mid \text{do}(A=a)]$ (CDT, severs the action's causes) vs. $\mathbb{E}[U \mid A=a]$ (EDT, keeps them as evidence).



EDT mistakes evidence for control

The smoking lesion

- A genetic lesion L causes cancer (cost C) and makes you more likely to want to smoke. Smoking itself is harmless and gives benefit $b > 0$.
- Conditioning on "I smoke" raises $P(L)$, hence raises estimated cancer risk: $\mathbb{E}[U \mid \text{smoke}] = b - C P(L \mid \text{smoke})$. For strong correlation, EDT **abstains** – even though abstaining does *nothing* to reduce cancer. EDT confuses a **symptom** for a **lever**.
- **Scorecard:** CDT and EDT are each half right.
 - CDT ignores predictive dependence $\Rightarrow$ loses Newcomb.
 - EDT overreacts to mere correlation $\Rightarrow$ loses the smoking lesion.
- The missing ingredient is a dependence that is neither raw physical causation nor raw evidential correlation.



Spurious counterfactuals: EDT breaks for self-aware agents

Conditioning on a probability-zero action

- A capable embedded agent can reason about (even *prove* things about) its own source code, so it can largely predict its own action. But then, for the action a it will *not* take, $P(A = a) = 0$.
- EDT's recipe $\mathbb{E}[U \mid A = a]$ is then $\frac{0}{0}$ – **undefined**. You may assign the unchosen action *any* value you like: a **spurious counterfactual**.
- So conditioning gives *no constraint* on counterfactual actions you are certain you won't take – exactly the cases decision-making turns on.
- (Foreshadow: in proof-theoretic form this same pathology becomes the **5-and-10 problem**, and the fix needs a notion of *logical* dependence.)



Functional decision theory: intervene on the function

The three rules side by side (Yudkowsky & Soares 2017)

- Write V for the outcome's value, Act for the physical-action node, and $\text{fdt}(P, G, x)$ for the *logical node* representing the output of the agent's decision function on observation x :

$$\text{EDT}(P, x) = \arg \max_a \mathbb{E}[V \mid \text{Obs}=x, \text{Act}=a],$$

$$\text{CDT}(P, G, x) = \arg \max_a \mathbb{E}[V \mid \text{do}(\text{Act}=a), \text{Obs}=x],$$

$$\text{FDT}(P, G, x) = \arg \max_a \mathbb{E}[V \mid \text{do}(\text{fdt}(P, G, x)=a)]$$

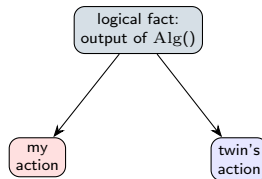
- The only change from CDT is *which node you intervene on*. CDT intervenes on the physical act – Peirce's **token** (“my action is a ”). FDT intervenes on the **type** – the abstract function – so every predictor, copy, or simulation computing that same function updates with it.
- Newcomb**: setting fdt =one-box also moves Ω 's earlier prediction $\Rightarrow$ box full; FDT one-boxes. **Smoking lesion**: the lesion does not compute your decision, so $\text{do}(\text{fdt}=\text{smoke})$ leaves it untouched; FDT smokes. **FDT gets both right**.
- Caveat: the function's output is fixed of *logical necessity*, so FDT must evaluate *counterpossibles* (antecedents that run counter-to-logic) – still the main open problem (cf. the 5-and-10 problem).



Twin Prisoner's Dilemma & subjunctive dependence

Same function, two bodies

- You and a psychological *twin* (your exact copy) play a one-shot Prisoner's Dilemma in separate rooms, no communication. You run the *same* decision function.
- **CDT**: "I can't causally affect her; defection dominates." Both defect, each gets \$1.
- **FDT**: "We compute the same function. If it outputs *C*, both cooperate; if *D*, both defect." So FDT cooperates – both get \$3.
- **Subjunctive dependence** (Y&S): when two physical systems compute the *same* function, their outputs are logically tied – like two calculators evaluating $6288+1048$: see one print 7336 and you know the other will too. Causal dependence is a *special case*; mere correlation (both calculators happen to be pink) is not. FDT acts as if it controls *all* instances of the computation at once.



one logical cause, two physical effects: CDT sees no arrow between them, FDT sees the shared parent



Counterfactual mugging: the updateless move

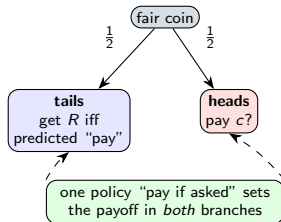
Paying in a branch you know you're not rewarded in

- Ω flips a fair coin. **Tails:** you get $R = \$10,000$ iff Ω predicted you'd pay on heads. **Heads:** Ω asks for $c = \$100$. You are asked to pay (so: heads). Pay?
- **Updateful** reasoning conditions on "I'm in heads": it sees only the \$100 cost and **refuses**. But *ex ante* the policy "pay if asked" is worth $\frac{1}{2}(R - c) > 0$.
- **UDT 1.0:** evaluate the local output against the **prior** – keep the (tails) branch whose payoff your policy controls, by dropping the updateful rule's extra $O=o$ conditioning:

$$\text{choice}(o) = \arg \max_a \mathbb{E}[U \mid \text{choice}(o)=a].$$

So UDT 1.0 **pays**.

- Updatelessness is *not* ignorance (no predictor $\Rightarrow$ UDT = ordinary EU max), and it is *reflectively stable*: the agent already follows the policy it would have precommitted to.





UDT 1.1: optimise the whole policy, not one action

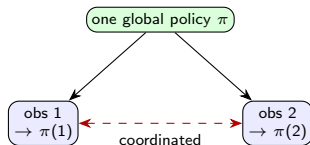
When local outputs must be coordinated

- Even *updateless local-action* selection can be too local. **Copied-agent anti-coordination**: two copies see index 1 or 2, must pick *different* actions to win (10 each), else 0.
- The local question “if $\text{Alg}(1) = A$, what is $\text{Alg}(2)$?” is the *wrong type*: both symmetric answers fail. The good object is the *whole table*
 $\text{Alg} : \{1, 2\} \rightarrow \{A, B\}$.
- **UDT 1.1 (Wei Dai)**: first choose the entire *policy* (a map $S : O \rightarrow A$) against the prior, *then* apply S^* to your actual observation o :

$$S^* \in \arg \max_{S: O \rightarrow A} \mathbb{E}[U \mid \text{choice}=S].$$

(UDT 1.0 instead fixes a single *output* and hopes the rest fall out right.)

- π_{AB}, π_{BA} score 10; a shared tie-breaker coordinates the copies. This is **global optimisation over branches**, not local optimisation.





The three shifts in “what am I choosing?”

CDT/EDT → FDT → UDT 1.0 → UDT 1.1 (after Kosoy's formalism)

Rule	Object chosen	Update status	Dependence used
CDT	local physical act	updateful	causal intervention
EDT	local act (as evidence)	updateful	evidential correlation
FDT / TDT	local algorithm output	updateful	logical / subjunctive
UDT 1.0	local algorithm output	updateless	logical / subjunctive
UDT 1.1	whole policy	updateless	logical over policies

- **Shift 1 (FDT):** from a physical act to the output of an abstract computation – captures predictors, copies, simulations.
- **Shift 2 (UDT 1.0):** from posterior to prior – keep the value of being predictable, don't discard branches whose payoff your policy controls.
- **Shift 3 (UDT 1.1):** from a single output to a whole policy – coordinate behaviour across all possible observations.



Problem 1: paying for worlds that don't exist

Ex-ante optimality is only as good as the prior

- UDT maximises *prior* expected utility. In the branch it actually inhabits it will **knowingly burn real utility**. In counterfactual mugging, after seeing heads, UDT hands over a real \$100 in the only world it will ever experience, to benefit a tails-world *it now knows did not happen*.
- Generalise: a UDT agent will take actions that are bad *in the actual world* because they would have been good in counterfactual scenarios that are never realised.
- This is dangerous when the prior is even slightly wrong about *which worlds are possible*: put prior mass on a predictor that does not actually exist (or on a logically impossible scenario), and UDT pays a **real, ongoing tax** for fake benefits. A clever (or merely hypothetical) Ω can extract resources by promising/threatening in branches that never occur.
- The worry: is “best from the prior” the right notion of rational, when “the prior” ranges over worlds you have since learned are not actual?



Problem 2: logical updatelessness

Which prior, when some of your uncertainty is mathematical?

- A real embedded agent's uncertainty is not only *empirical* (coin flips). It is also **logical**: it does not know all consequences of its own source code, the 10^{100} -th digit of π , or whether the opponent's program halts.
- UDT says "don't update." Refusing to update on *empirical* observations is clear. Refusing to update on *logical* facts is not: as the agent thinks longer it *learns* logical facts (proves theorems, computes digits) – and that looks exactly like updating.
- **Logical updatelessness** would have the agent choose its policy as if from a state of *logical* ignorance, not taking computed logical facts into account. But (a) it cannot help computing some of them, and (b) we have **no coherent "logical prior"** – a probability distribution over mathematical truths before any computation.
- This is open. A satisfactory UDT needs a theory of *logical uncertainty* (cf. logical induction), which we do not yet have. It is also the doorway to the 5-and-10 problem.



The 5-and-10 problem

The simplest decision, made hard by self-knowledge (Demski & Garrabrant)

- You may take \$5 or \$10; utility = the dollars. A child takes \$10. Can a fully self-aware embedded agent?
- Frame it for an embedded agent. The world is a deterministic program that pays out whatever number you take; the agent has *its own source code* and the world's. It chooses by **proof search**: look for proofs of " $(A=\$5) \Rightarrow U=x$ " and " $(A=\$10) \Rightarrow U=y$ ", and take the action with the higher *proven* payoff.
- **The deep issue.** "If you can know your own behaviour, it becomes hard to reason about what would happen if you behaved *differently*." Because the agent is deterministic it takes exactly one action; the other is ruled out *by logic*, so "what if I did otherwise?" is a counterfactual with an **impossible antecedent**.
- **The trap, concretely.** Suppose the agent can prove it takes \$5. Then $A \neq \$10$, so the material conditional " $(A=\$10) \Rightarrow (U=\$0)$ " is **vacuously true** (a false antecedent implies anything). The agent proves it, "concludes" \$10 is worthless, and takes \$5 – *confirming* the proof. A self-fulfilling **spurious counterfactual**.
- The fault is not bad logic but *too much self-knowledge*: the action is a fixed fact, and a naive counterfactual over a fixed fact is vacuous.



Spurious proofs and Löb's theorem

Why a naive proof-searcher is genuinely unsafe

- **Löb's theorem (formula + intuition).** Write $\Box P$ for " P is provable". Then

if $\vdash (\Box P \rightarrow P)$ then $\vdash P$; equivalently $\vdash \Box(\Box P \rightarrow P) \rightarrow \Box P$.

Intuition: a system may treat "provable $\Rightarrow$ true" as safe *only for what it already proves* – so strongly that it cannot even prove its own consistency ($\Box \perp \rightarrow \perp$, i.e. Gödel II).

- **The spurious proof is available.** Let $S := ((A=\$10) \Rightarrow (U=\$0))$. The agent is built so that *proving* S makes it take \$5; then $A \neq \$10$, so S holds vacuously, giving $\vdash (\Box S \rightarrow S)$. By Löb, $\vdash S$. A naive proof search can therefore prove the bad statement and take \$5.
- **Moral.** Ruling out spurious proofs needs extra structure (e.g. "playing chicken with the universe"); a general account of *logical counterfactuals* (counterpossibles) is open. The same Löbian obstacle blocks an agent from fully trusting a successor's proofs – a reflective-stability problem.



Why we want a formal optimality criterion

Beyond case-by-case scoring

- Newcomb, counterfactual mugging, the 5-and-10 problem, the twin PD, open-source games all share a **common structure**: agent and environment are both *programs* that may inspect each other's code, simulate each other's behaviour, or contain subroutines algorithmically *isomorphic* to one another.
- Yet these are usually presented as standalone puzzles, and CDT/EDT/FDT/UDT are judged *case by case* ("does it get Newcomb right?") rather than against a single optimality standard.
- **The substantive question**: can we define one class of computational environments rich enough to contain all these problems at once, together with a criterion of optimality that any candidate decision theory can be *measured* against?



Agents and worlds as programs

The setup

- **Agent** = a program with access to (i) its own source code and (ii) the world's source code; it returns an action.
- **World** = a program that may
 - *call the agent* as a function (input $\rightarrow$ action), possibly *many times on different inputs* ("what would the agent do if it observed x ?"), and
 - *search for proofs* about the agent's input-output behaviour;it returns a utility.
- **Goal:** design an agent whose utility is high across (almost) all worlds – the best mapping from situations to actions.





Fairness

Judging the agent only on its behaviour

- Without a constraint, a “world” could just punish source code: “if the agent is written in Python, utility = 0.” Then *no* decision theory could do well and the criterion would be vacuous.
- **Fairness:** the world may depend on the agent *only through its input–output behaviour* (its policy), never through implementation details unrelated to its outputs. Equivalently, a fair world is representable as a program that takes the agent's *policy/behaviour* as input.
- Newcomb, the smoking lesion, counterfactual mugging, the twin PD, and the open-source PD are all **fair**: the predictor/lesion/copy interacts with the agent through its behaviour. This is what makes them legitimate tests.



What a good agent must discover

Two demands – which are exactly FDT/UDT turned into a spec

- **Coordinate across function calls.** If the world queries the agent on several inputs (probing its counterfactual behaviour), the agent must *recognise this* and keep its counterfactual outputs *coordinated* for high utility. (This is precisely UDT 1.1's move to optimise a whole policy.)
- **Subjunctive dependence.** If the world contains a subroutine *algorithmically isomorphic* to the agent (a copy in different code), the agent should treat its own decision as also fixing that subroutine's output – and so must *detect* that “the world contains something isomorphic to me.” (This is precisely FDT.)

The open problem (and the bridge ahead)

- Define the class of fair computational worlds precisely; define optimality (e.g. dominance across all fair worlds); show CDT/EDT/FDT/UDT differ on it; and ask whether any (or some modification) is *optimal*.
- Caveats already met: logical counterfactuals (5-and-10), logical uncertainty / logical updatelessness, what counts as a “policy”, and multi-agent fairness & bargaining – which lead into reflective oracles, open-source games, and commitment races.